BUSINESS PROBLEM STATEMENT — MASTERCONTROL LEAD SCORING

Course: MSBA IS 6813 Capstone | **Spring 2026** | University of Utah

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Sponsor: MasterControl

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1. BUSINESS PROBLEM

- MasterControl provides quality and manufacturing management software to life sciences companies — pharma, biotech, and medical device manufacturers
- They sell **two** core product lines:
  - **Qx** (**Quality Solutions**) — established product, **20+** years in market
  - **Mx** (**Manufacturing Solutions**) — newer product, **~4** years in market
- Leads enter the sales pipeline as **Qualified Activity Leads** (**QALs**) and are worked by **Sales Development Representatives** (**SDRs**) who attempt to progress them through stages: **Recycled**, **Excluded**, **SQL**, **SQO**, or **Won**
- The core problem: **Mx** converts **QALs** to **SQL** at **12.7%** versus **19.7%** for **Qx**
  - That is a **7** percentage-point gap — **Qx** outperforms **Mx** by **55%**
  - This represents significant unrealized pipeline value for a product MasterControl is actively investing in growing
- The sponsor believes the gap may stem from targeting — **Mx** may be getting pitched to the wrong types of companies or the wrong people within those companies, rather than reflecting a product quality issue
- **SDRs** have limited daily call and email capacity. They currently lack a systematic, data-driven method for deciding which leads to contact next. Per the sponsor: the goal is a ranking system, not just a list
- The dataset contains **16,644 QAL** records spanning **January 2024** through **December 2025** (**two** complete years), with **15,371** unique contacts across **14** fields
- Overall, **18.0%** of leads progressed (**2,988** of **16,644**). But this masks the **Mx**/**Qx** split — and understanding what drives that split is the central question

2. BENEFIT OF A SOLUTION

- **Prioritized prospecting:** A lead scoring model would give **SDRs** a ranked queue so the highest-probability **Mx** leads get called first, not last. This directly addresses the capacity constraint the sponsor identified

- **Refined Ideal Customer Profile** (**ICP**): By identifying which industries, manufacturing models, company sizes, and job titles are associated with **Mx** conversion, MasterControl can sharpen its targeting and stop spending **SDR** time on low-probability segments
- **Channel efficiency:** The dataset includes **12** distinct marketing channels (**Online Ads**, **Email**, **SEO**, **Direct/Inbound**, **External Demand Gen**, etc.). Understanding which channels produce leads that actually convert — versus which just produce volume — could inform marketing budget allocation
- **Recoverable pipeline: 72.2%** of all **QALs** are recycled back to marketing. If even a fraction of those recycled leads are viable prospects that simply stalled, a scoring model could surface the ones worth re-engaging
- **Cross-sell identification:** Some account/title combinations may convert well for both **Mx** and **Qx**. Identifying these "**universal**" profiles gives the sales team a natural expansion playbook

3. SUCCESS METRICS

- **Primary model metric: AUC-ROC** — this measures how well the model ranks leads by conversion probability. We prefer **AUC** over raw accuracy because with an **82%** negative class rate, a model that predicts "no conversion" for every lead would be **82%** accurate and completely useless
- **Calibration quality:** The model's predicted probabilities should be meaningful — a lead scored at **30%** should convert roughly **30%** of the time. We will evaluate this with calibration curves and **Brier Score**
- **Business-level success:** Did the model-recommended subset of **Mx** leads convert at a meaningfully higher rate than the current **12.7%** baseline? The sponsor cares mostly about conversion rate lift
- **Profit-based evaluation:** Using the sponsor's business parameters (cost / sales call vs. value per **SQL** conversion), we will find the scoring threshold that maximizes net dollar impact — the point where contacting additional leads stops being profitable
- **Actionability:** The final deliverable must translate into a concrete recommendation the sales team can execute. A model that performs well statistically but cannot answer "who should I call next?" has not succeeded

4. ANALYTICS APPROACH

- **Framework: CRISP-DM** — we will follow the standard cycle of **Business Understanding**, **Data Understanding**, **Data Preparation**, **Modeling**, **Evaluation**, and **Deployment** recommendations
- **Target variable:** We will construct a binary outcome where **SQL**, **SQO**, and **Won** = **1** (success) and all other stages = **0**.

	Predicted Class			
		SUCCESS/POSITIVE SQL, SQO, and Won = 1	**FAILURE/NEGATIVE** All other stages = 0	
Actual Class	**SUCCESS/POSITIVE** SQL, SQO, and Won = 1	**True Positive (TP)** = Our model predicted the lead to progress and the lead in fact did progress	**False Negative (FN)** = **Type II Error =** Our model predicted the lead to not progress when in reality the lead did progress	**Sensitivity** = TP / (TP + FN)
	FAILURE/NEGATIVE All other stages = 0	**False Positive (FP)** = **Type I Error =** Our model predicted the lead to progress when in fact the lead did not progress	**True Negative (TN)** = Our model predicted the lead to not progress and the lead in fact did not progress	**Specificity** = TN / (TN + FP)
		Precision = TP / (TP + FP)	**Negative Predictive Value** = TN / (TN + FN)	**Accuracy** = (TP + TN) / (TP + TN + FP + FN)

- The dataset is primarily categorical (**13** of **14** fields are text-based), which shapes the analytical approach:
  - Account-level features are well-structured with low missingness (<**1%** for industry, manufacturing model, site function, tier, and territory). These can be analyzed directly
  - The contact/lead title field is free-text with **~4,000** unique values and **29%** missing. This will require **NLP** preprocessing — parsing titles into structured dimensions like seniority level (**C-Suite**, **VP**, **Director**, **Manager**), functional area (**Quality**, **Operations**, **Manufacturing**, **Regulatory**), and keyword flags
  - The priority field contains **11** levels that encode lead intent — from high-intent actions like "**P1 - Contact Us**" and "**P1 - Website Pricing**" to lower-signal categories like "**Priority 2**" content downloads. The sponsor confirmed this distinction matters
  - The marketing channel field (**12** levels) likely contains signal about lead quality — inbound/organic channels may produce higher-intent leads than outbound/paid channels
- Given the categorical-heavy feature space, an **~18%** positive class rate, and the need for interpretable output, we anticipate evaluating:
  - Tree-based ensemble methods (gradient boosting variants, random forests) — these handle categorical features natively, manage nonlinear relationships, and support

feature importance analysis
    - **Logistic regression** — as a baseline and for its natural probability calibration and coefficient interpretability
    - **Ensemble/stacking** approaches — if individual models show complementary strengths
- **Explainability is not optional.** The sponsor needs to understand **WHY** a lead scores high — not just that it does. We plan to use techniques like **SHAP** values to decompose individual predictions into feature-level contributions
- **Two-level analysis structure** (per the data dictionary guidance):
    1. **Account-level**: Which industries, manufacturing models, site functions, and company sizes are best for **Mx**?
    2. **Title-level**: Within those account types, which job titles and seniority levels convert best?
    3. **Interaction effects**: Do certain account + title combinations outperform what either dimension would predict alone?

5. SCOPE

IN SCOPE:

- Binary classification model predicting **QAL-to-SQL** conversion
- Lead scoring system that ranks leads by predicted conversion probability
- Exploratory analysis of the **Mx** vs. **Qx** conversion gap — what account types, titles, channels, and priority levels explain it
- Feature importance analysis translating model internals into actionable targeting recommendations
- Profit curve analysis quantifying the dollar impact of model-guided lead prioritization
- **Cross-sell** analysis: which profiles convert well for both products
- Final deliverables: presentation to MasterControl, group **EDA** and modeling notebooks, **GitHub** portfolio

OUT OF SCOPE:

- Predicting deal size or revenue — the sponsor confirmed the goal is conversion rate optimization, not profitability per deal
- Existing customer upsell — all **16,644** records are net-new prospects
- Real-time **CRM** integration or production deployment of a live scoring model
- Causal inference on why leads convert — this is a predictive modeling engagement, not an experimental design

FUTURE EXTENSIONS (could be added if time permits or in a follow-on):

- Integration with MasterControl's existing internal **AI** ranking system (which currently scores accounts and contacts **0-100**)

- Enrichment with third-party data sources (**ZoomInfo**, **Clearbit**) to fill gaps — notably the **14.6%** of accounts with "**Not Enough Info**" for manufacturing model and the **29%** missing titles
- Longitudinal tracking of recycled lead re-engagement outcomes
- Regional go-to-market recommendations (**APAC** underperforms per the sponsor, possibly due to weaker regulatory environment)

6. DETAILS

GENERAL TIMELINE:

- Business Problem Statement - Feb 1st
- EDA Group Notebook - Feb 18th
- Modeling Notebook - Mar 18th
- Practice Presentation - Apr 5th
- Final Sponsor Delivery - Apr 8th
- Portfolio & Peer Eval - Apr 19th